RV College of Engineering®

Go, Change the World

Academic Year 2023-24 (EVEN Semester)

USN	1	R	V							

Department of Artificial Intelligence and Machine Learning

Course Code:	:	AI244AI	Date	:	23-07-2024
Semester	:	IV	Time	:	12:00 – 1:30 PM
Max Marks	:	50	Duration	:	90 mins

Artificial Intelligence and Machine Learning

CIE II

Note: Answer all the Questions

SL. No		Questions	M	BT	CO
1	a	Discuss the five components of a well-defined AI problem. Ans: (Each carries 1 M with a brief description and examples) 1. Initial State 2. Actions 3. Transition table/action description 4. Goal Test 5. Path Cost	5	2	CO1
	b	Give a well-defined problem definition for the 8-Queens problem. Ans: (Each carries 1 M with a brief description) 1. **States**: Any arrangement of 0 to 8 queens on the board is a state. 2. **Initial state**: No queens on the board. 3. **Actions**: Add a queen to any empty square. 4. **Transition model**: Returns the board with a queen added to the specified square. 5. **Goal test**: 8 queens are on the board, none attacked	5	2	CO5
2	a	Discuss the application of Breadth First Search (BFS) in a travel route planning application. Ans: 1. Use of the graph data structures for modeling the locations map (1M) 2. Defining the Goal State (1M) 3. BFS Algorithm: (3M) a. Use of Queues to store the locations b. Maintaining the visited locations information c. Start from a given location in the front of the queue d. Check if this is the required location. If yes, the Goal is reached e. If not, enqueue the reachable locations from the given location f. Mark the current location has been visited and repeat the steps until all the locations available on the map are explored	5	3	CO2

<table>
<tr><td></td><td>b</td><td>Differentiate between Depth First Search (DFS) and Depth-limited Search (DLS). Web crawling systematically searches the World Wide Web to index and collect website data. Which of the two DFS and DLS suits this purpose, and why?

Ans: Any 2 differences like, working principles of searching techniques and the time/space complexity. (2M)
DLS is suitable (1M)
Correct justification like avoiding infinite crawling (2M)</td><td>5</td><td>3</td><td>CO2</td></tr>
<tr><td rowspan="2">3</td><td>a</td><td>For the given data set shown in Table 3a., find whether (20,35) belongs to class Red or Blue using the KNN neighbor. Assume K = 5.

Table 3a
<table>
<tr><td>Brightness</td><td>Saturation</td><td>Class</td></tr>
<tr><td>40</td><td>20</td><td>Red</td></tr>
<tr><td>50</td><td>50</td><td>Blue</td></tr>
<tr><td>60</td><td>90</td><td>Blue</td></tr>
<tr><td>10</td><td>25</td><td>Red</td></tr>
<tr><td>70</td><td>70</td><td>Blue</td></tr>
<tr><td>60</td><td>10</td><td>Red</td></tr>
<tr><td>25</td><td>80</td><td>Blue</td></tr>
<tr><td>20</td><td>35</td><td>????</td></tr>
</table></td><td>5</td><td>L3</td><td>CO1</td></tr>
<tr><td>b</td><td>For the given dataset shown in table 3b. (use logistic regression)

Table 3b.
<table>
<tr><td>Hours Study</td><td>Pass (1)/Fail (0)</td></tr>
<tr><td>29</td><td>0</td></tr>
<tr><td>15</td><td>0</td></tr>
<tr><td>33</td><td>1</td></tr>
<tr><td>28</td><td>1</td></tr>
<tr><td>39</td><td>1</td></tr>
</table>
1. Given the class, how can we calculate the probability of passing for the student who studied 33 hours?
2. At least how many hours a student should study to ensure he will pass the course with a probability of > 95%.
3. Assume the model suggested by the optimizer for the odds of posing the course is log(odds) = -64+2*hours.</td><td>5</td><td>L3</td><td>CO2</td></tr>
<tr><td>4</td><td>a</td><td>Consider the given dataset, apply the naïve Bayes algorithm, and predict that if a fruit has the following properties, then which type of the fruit it is Fruit = {Yellow, sweet, long}

Frequency table:
<table>
<tr><td>Fruit</td><td>Yellow</td><td>Sweet</td><td>Long</td><td>Total</td></tr>
<tr><td>Mango</td><td>350</td><td>450</td><td>0</td><td>650</td></tr>
<tr><td>Banana</td><td>400</td><td>300</td><td>350</td><td>400</td></tr>
</table></td><td>06</td><td>L4</td><td>CO5</td></tr>
</table>

			Others	50	100	50	150				
			Total	800	850	400	1200				
	b	With neat sketches, explain Bagging and Boosting ensemble learning techniques.							04	L2	CO1
5		Consider a training dataset of 6 data instances shown in Table 5. Use four decision stumps for each of the four attributes. Apply the AdaBoost algorithm and classify the dataset with job offers as the target attribute.							10	L4	CO5

Table 5

CGPA	Interactiveness	Practical Knowledge	Comm skills	Job profile
>=9	Yes	Good	Good	Yes
<9	N	G	Moderate	Y
>=9	N	Avg	M	N
<9	N	Avg	G	N
>=9	Y	G	M	Y
>=9	Y	G	M	Y

M-Marks, BT-Blooms Taxonomy Levels, CO-Course Outcomes

Marks Distribution	Particulars	CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
	Max Marks CIE	14	15	-	-	21	-	14	20	16	-	-

Course Outcomes: After completing the course, the students will be able to:-	
CO1	Explain and apply AI & ML algorithms to address various requirements of real-world problems.
CO2	Design and develop AI and ML solutions to benefit society, science, and industry.
CO3	Use modern tools to create AI and ML solutions.
CO4	Demonstrate effective communication through team presentations and reports to analyze the impact of AI and ML solutions on society and nature.
CO5	Conduct Performance evaluation, modeling, and validation of AI and ML solutions benefitting lifelong learning.

Q. No	Scheme and Solutions	M
3a.	d1= (√ (20-40) ²+(35-20) ²) =√ (400+225) = √625 = 25, d2=√ ((20-50) ²+(35-50) ²) = √ (900+225) =√1125= 33.54, d3=√ ((20-60) ²+(35-90) ²) =√ (1600+3025) =√4625 = 68.01	05

BRIGHTNESS	SATURATION	CLASS	DISTANCE
40	20	Red	25
50	50	Blue	33.54
60	90	Blue	68.01
10	25	Red	10
70	70	Blue	61.03
60	10	Red	47.17
25	80	Blue	45

Let's rearrange the distances in ascending order:

BRIGHTNESS	SATURATION	CLASS	DISTANCE
10	25	Red	10
40	20	Red	25
50	50	Blue	33.54
25	80	Blue	45
60	10	Red	47.17
70	70	Blue	61.03
60	90	Blue	68.01

Since we chose 5 as the value of **K**, we'll only consider the first five rows. That is:

BRIGHTNESS	SATURATION	CLASS	DISTANCE
10	25	Red	10
40	20	Red	25
50	50	Blue	33.54
25	80	Blue	45
60	10	Red	47.17

As you can see above, the majority class within the 5 nearest neighbours to the new entry is **Red**. Therefore, we'll classify the new entry as **Red**.

3b.

1) if <0.5 – Student will fail
>0.5 – " " pass

→ 1) Calculate the probability of pass for the student who studied 33 hours.

Sol $S(x) = \frac{1}{1+e^{-x}}$ Interms of probability we write it as $p = \frac{1}{1+e^{-z}}$

log (odds) = $Z = -64 + 2 * \text{hours}$

$= -64 + 2(33)$

$= -64 + 66 = 2$

$$\boxed{p(x) = \frac{1}{1+e^{-2}} = 0.88}$$

∴ Inference: if student studies for 33 hrs, then there is a chance that the student will pass the exam.

2) No. of hours? a student has to study so that he passes the exam with probability of >95%.

Sol $p = \frac{1}{1+e^{-z}} = 0.95 = \frac{1}{1+e^{-z}} = 0.95$

Simply it

$0.95 * (1+e^{-z}) = 1$

$0.95 * e^{-z} = 1 - 0.95$

$0.95 + 0.95e^{-z} = 1$

$0.95e^{-z} = 1 - 0.95$

$0.95e^{-z} = 0.05$

$e^{-z} = \frac{0.05}{0.95}$

$= 0.0526$

$e^{-z} = 0.0526$ $\ln(e^{x}) = x$

$\ln(e^{-z}) = \ln(0.0526)$

$-z = \ln(0.0526) = -2.94$

$z = 2.94$

log (odds) = $Z = -64 + 2 * \text{hrs}$

$2.94 = -64 + 2 * \text{hrs}$

$2.94 + 64 = 2 * \text{hrs}$

$2 * \text{hrs} = 66.94$

$\text{hrs} = \frac{66.94}{2} = 33.47 \text{ Hrs}$

Inference: the student should study at least 33.47 hours, so that he will pass the exam with >95%.

05

4a. (06)

Solution: P(A|B) = (P(B|A) * P(A))/ P(B)

1. Mango: P(X | Mango) = P(Ye | Yellow) * P(Sweet | Mango) * P(Long | Mango)

a)P(Yellow | Mango) = (P(Mango | Yellow) * P(Yellow))/ P (Mango) = ((350/800) * (800/1200)) / (650/1200)

P(Yellow | Mango)= 0.53 →1

1.b)P(Sweet | Mango) = (P(Sweet | Mango) * P(Sweet))/ P (Mango) = ((450/850) * (850/1200)) / (650/1200)

P(Sweet | Mango)= 0.69 → 2

1. c)P(Long | Mango) = (P(Long | Mango) * P(Long))/ P (Mango) = ((0/650) * (400/1200)) / (800/1200)

P(Long | Mango)= 0 → 3

On multiplying eq 1,2,3 ==> P(X | Mango) = 0.53 * 0.69 * 0

P(X | Mango) = 0

2. Banana: P(X | Banana) = P(Yellow | Banana) * P(Sweet | Banana) * P(Long | Banana)

2.a) P(Yellow | Banana) = (P(Banana | Yellow) * P(Yellow))/ P (Banana) = ((400/800) * (800/1200)) / (400/1200)

P(Yellow | Banana) = 1 → 4

2.b) P(Sweet | Banana) = (P(Banana | Sweet) * P(Sweet))/ P (Banana) = ((300/850) * (850/1200)) / (400/1200)

P(Sweet | Banana) = .75→ 5

2.c)P(Long | Banana) = (P(Banana | Yellow) * P(Long))/ P (Banana) = ((350/400) * (400/1200)) / (400/1200)

P(Long | Banana) = 0.875 → 6

On multiplying eq 4,5,6 ==> P(X | Banana) = 1 * .75 * 0.875

P(X | Banana) = 0.6562

P(X|Mango) 2 M, P(X|Banana) 2 M, P(X|Others) 2 M

3. Others: P(X | Others) = P(Yellow | Others) * P(Sweet | Others) * P(Long | Others)

3.a) P(Yellow | Others) = (P(Others| Yellow) * P(Yellow))/ P (Others) = ((50/800) * (800/1200)) / (150/1200)

P(Yellow | Others) = 0.34→ 7

3.b) P(Sweet | Others) = (P(Others| Sweet) * P(Sweet))/ P (Others) = ((100/850) * (850/1200)) / (150/1200)

P(Sweet | Others) = 0.67 → 8

3.c) P(Long | Others) = (P(Others| Long) * P(Long))/ P (Others) = ((50/400) * (400/1200)) / (150/1200)

P(Long | Others) = 0.34 → 9

On multiplying eq 7,8,9 ==> P(X | Others) = 0.34 * 0.67* 0.34

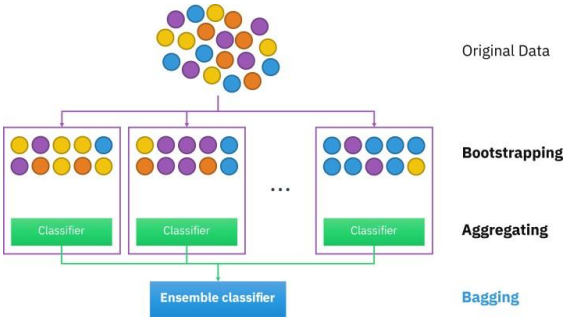
P(X | Others) = 0.07742

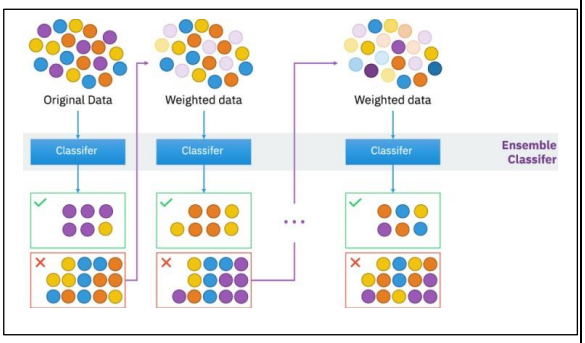
So finally from P(X | Mango) == 0 , P(X | Banana) == 0.65 and P(X| Others) == 0.07742. **We can conclude Fruit{Yellow,Sweet,Long} is Banana.**

4b. (04)

Bagging (Bootstrap AGGregatING)

- Bootstrap method is used to create multiple subsets of training data set and then train multiple base models separately on these bootstrapped training set to get different predictions from these base models.
- The final prediction is obtained by averaging or voting on the predictions from these base models.





Boosting:

- Boosting uses the entire training data set in a sequential manner to improve the performance of weak learners.
- First, using the whole training data set to fit on a weak leaner.
- Then, according to the prediction results to adjust the weight of each observation in the training data set.
- Giving lower weights to those were classified correctly, while giving higher weights to those were classified incorrectly.
- Fitting a new weak learner on this modified data set.

	• The sequential process of adjusting weights and predictions continues until a stopping criterion is reached. • The final prediction is a weighted result of all weak learners.	
5	1st decision stump classify the instances based on CGPA attribute CGPA: >=9, <9, >=9, <9, >=9, >=9 Predicted Job offer: Y, N, Y, N, Y, Y Actual Job: Y, Y, N, N, Y, Y weight: 1/6, 1/6, 1/6, 1/6, 1/6, 1/6 (NM) (NM) (M) (M) (M) (M) CGPA Int: 1 9 Y; 2 9 M; 3 A N; 4 A G; 5 M Y; 6 N Y So whenever CGPA >= 9 → Job offer predicted = 'Yes' otherwise 'NO' Step:2 Compute the weighted error ε_{CGPA} of H_{CGPA} on current training dataset T. $\varepsilon_i = \sum_{j=1}^{N} H_i(d_j) \cdot wt(d_j)$ (hypothesis of jth example; weight of j sample) where $H_i(d_j) = 0$ for correct prediction, $= 1$ for wrong prediction $\varepsilon_{CGPA} = [0 \times \frac{1}{6} + 1 \times \frac{1}{6} + 1 \times \frac{1}{6} + 0 \times \frac{1}{6} + 0 \times \frac{1}{6} + 0 \times \frac{1}{6}] = 2 \times \frac{1}{6} = \frac{1}{3} = 0.333$ Step:2(c) Compute the weight of each weak classifier $\alpha_{CGPA} = \frac{1}{2} \ln \frac{(1 - \varepsilon_{CGPA})}{\varepsilon_{CGPA}} = \frac{1}{2} \ln \frac{(1 - 0.333)}{0.333} = 0.347$ Step: 2(d) Calculate the normalization factor Z_{CGPA} (weight of weak classifier) $Z_{CGPA} = wt(\text{correct classified instance}) * e^{-\alpha_{CGPA}}$ (no. of correct classification) $+ wt(\text{wrong classified instance}) * e^{+\alpha_{CGPA}}$ (no. of wrong classification) $= \left(\left(\frac{1}{6} \times 4\right) \cdot e^{-0.347}\right) + \left(\left(\frac{1}{6} \times 2\right) \cdot e^{0.347}\right) = 0.9428$ Step:2(e) update the weights of all data instances $wt(d_j)_{i+1} = \frac{wt(d_j)_{CGPA} \text{ of correct instance} \cdot e^{-\alpha_{CGPA}}}{Z_{CGPA} - (\text{normal factor})}$ $wt(d_j)_{i+1} = \frac{\frac{1}{6} * e^{-0.347}}{0.9428} = 0.1249$ (correct classification weight) $wt(d_j)_{i+1} = \frac{wt(d_j)_{CGPA} \text{ of incorrect instances} \cdot e^{\alpha_{CGPA}}}{Z_{CGPA}} = \frac{\frac{1}{6} \times 2 \times e^{0.347}}{0.9428} = 0.2501$ (incorrect classification weight) ⇒ II Decision stump for interactiveness Interactiveness: Y, N, N, N, Y, Y Predict: Y, N, N, N, Y, Y Actual Job: Y, Y, N, N, Y, Y weight: 0.1249, 0.2501, 0.2501, 0.1249, 0.1249, 0.1249 if interactiveness is 'yes' the data instance is predicted to have 'job offer' else 'NO' as 'yes' → Compute the weighted error $\varepsilon_{interact}$ of $H_{interact}$ $\varepsilon_i = \sum_{j=1}^{N} H_i(d_j) wt\, d(j)$ where $H_i(d_j) = 0$ for correct, $= 1$ for wrong. $\varepsilon_{interact} = 1 * 0.2501 = 0.2501$ Step 2(c) Compute the weight of each weak classifier $\alpha_{interact} = \frac{1}{2} \ln \frac{(1 - \varepsilon_{interact})}{\varepsilon_{interact}}$ $\Rightarrow \alpha_{interact} = \frac{1}{2} \ln \frac{(1 - 0.2501)}{0.2501} = 0.5490$ Step:2(d) Calculate the normalization factor. $Z_{interact} = wt(\text{correct classif Int}) * \text{no. of corr clas} * e^{-\alpha_{interact}} (+) wt(\text{incorrect class int}) * \text{no. of incorr clas} * e^{+\alpha_{interact}}$ $Z_{int} = 0.1249 * 4 * e^{-0.549} + 0.2501 * 1 * e^{-0.549} + 0.2501 * 1 * e^{0.549}$ $= 0.866$ (NF) Step: 2(e) update the weight of all data instances: $wt(d_j)_{i+1} = \frac{wt(d_j)_{int} \text{ of correct instances} * e^{-\alpha_{int}}}{Z_{interact}}$ $wt(d_j)_{i+1} = \frac{0.1249 * e^{-0.549}}{0.866} = 0.0832$ (for correct classification but weight changed) $wt(d_j)_{i+1} = \frac{0.2501 * e^{-0.549}}{0.866} = 0.166$ (3) 0.2501	10

$$wt(d_i)_{t+1} = \frac{wt(d_i)_{t} \text{ of incorr inst} \times e^{\alpha_{int}}}{Z_{int}}$$

(4)

$$wt(d_i)_{t+1} = \frac{0.2501 * e^{0.549}}{0.866} = 0.5001$$

intern	P job	Act	weight
Y	Y	Y	0.0832
N	N	Y	0.5001
N	N	N	0.1667
N	N	N	0.0832
Y	Y	Y	0.0832
Y	Y	Y	0.0832

iii. Decision stump for practical knowledge

Pk	P job	Act job	weight
Good	Y	Y	0.0832
G	Y	Y	0.5001
Avg	N	N	0.1667
Avg	N	N	0.0832
G	Y	Y	0.0832
G	Y	Y	0.0832

both r same & correctly classified

So no need to work on this. Now we will go for the last one.

iv Decision stump for Comm skill

2(a). train the DS H_{cs} with sample obtain in the previous weak class H_{cs}

Comm sk	P job	A job	weight
Good	Y	Y	0.0832
mod	N	Y	0.5001
mod	N	N	0.1667
Good	Y	N	0.0832
Mod	N	Y	0.0832
Mod	N	Y	0.0832

2(b) compute weighted error: e_{cs}

$$\varepsilon_i = \sum_{j=1}^{N} H_i(d_j)\, wt(d_j)$$

where $H_i(d_j) = 0$ for correct
$= 1$ for wrong.

$$e_{cs} = (1 * 0.5001 + 3 * 0.0832)$$

$$e_{cs} = 0.7497$$

Compute the weight of each weak classifier (5)

2(c) $\alpha_{cs} = \frac{1}{n} \ln \frac{(1-\varepsilon_{cs})}{\varepsilon_{cs}}$

$$\alpha_{cs} = \frac{1}{2} \ln \frac{(1-0.7497)}{0.7497} = -0.5485$$

$\Rightarrow$ 2(d) Ans

$Z_{cs} =$ wt (correct class inst) * no. of corr class $\times e^{-\alpha_{cs}}$ + wt (wrong class I) * no. of wrong class $\times e^{\alpha_{cs}}$

$$Z_{cs} = 0.0832 * 1 * e^{-(-0.5485)} + 0.1667 * 1 * e^{-(-0.5485)} + 0.5001 * 1 * e^{+(0.5485)} + 0.0832 * 3 * e^{+(0.5485)}$$

$$Z_{cs} = 0.866$$

2(e): Update the weights of all data instances

1. $$wt(d_j)_{i+1} = \frac{wt(d_i)_{cs} \text{ of corr inst} * e^{-\alpha_{cs}}}{Z_{cs}}$$

$$= \frac{0.0832 * e^{-(-0.5485)}}{0.866} = 0.1663$$

$$wt(d_j)_{t+1} = \frac{0.1667 * e^{-(-0.5485)}}{0.866} = 0.331$$

$$wt(d_j)_{t+1} = \frac{wt(d_i) \text{ of incorr class Ins} * e^{\alpha_{cs}}}{NF}$$

$$wt(d_j)_{t+1} = \frac{0.5001 \times e^{+(0.5485)}}{0.866} = 0.3337$$

$$wt(d_j)_{i+1} = \frac{0.0832 * e^{+(-0.5485)}}{0.866}$$

$$= 0.0555$$

Comm sk	Pred job	Act job	weight
Good	Y	Y	0.0832
mod	N	Y	0.3337
mod	N	N	0.3331
Good	Y	N	0.0555
mod	N	Y	0.0555
mod	N		0.0555

Step-3 Compute the final predicted value for each data instance:

S.no	$\alpha_{capA} = 0.347$	$\alpha_{int} = 0.5490$	$\alpha_{comm} = 0.5485$	W_{Avg} (6)	Final Pred
1	Y	Y	Y	0.3475	Y
2	N	N	N	0	N
3	Y	N	Y	0.347	Y
4	N	N	Y	−0.5485	N
5	Y	Y	N	0.896	Y
6	Y	Y	N	0.896	Y

$$H_f(d_j) = \sum_{i=1}^{n} \alpha_i * H_i(d_j)$$

$$H_f(d_j) = \alpha_{capA} * yes + \alpha_{int} * y + \alpha_{cs} * yes$$

$$= 0.347 * 1 + 0.5490 * 1 - 0.5485 * 1$$

$$= 0.3475$$

if yes = 1
No = 0
So 2nd ex = 0

if $w_{avg} >= 0 \rightarrow$ yes
else No.